

Pricing Risk at a Boundary

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October 9, 2026

Abstract

Economics prices the risk of crossing a tipping point through its probability. I show that this is not enough: when emissions bring a system such as an ocean circulation or a forest closer to tipping, the value of restraint depends on how fast the threshold approaches. Two systems whose probability of crossing rises at the same rate can carry prices that behave in opposite ways: one stays bounded as the remaining emissions budget runs out, the other does not. Observations cannot tell them apart in time. What tells them apart is the local geometry of the tipping point. At a fold, the generic case, the price rises as the inverse square root of the remaining budget once crossing is likely, and the benefit of restraint steepens faster than its cost, which favours a quantity over a price. That quantity is a date: the year after which even stopping emissions at once cannot avoid the threshold. It needs no damage estimate, and policy should use it alongside the carbon price.

JEL codes: Q54, Q58, H23, D81, C61.

Keywords: tipping points, fold bifurcation, carbon budget, prices versus quantities.

1 Introduction

When a controllable stock drifts toward an irreversible boundary, what is the marginal value of staying on the safe side? The question arises wherever spending a budget brings a volatile system closer to a point of no return: carbon emissions and the tipping points of the climate, leverage and default, harvesting and the collapse of a population. Economics answers it through the probability of crossing. It writes a hazard, a probability of crossing per unit of time that rises with the stock, attaches a loss to the crossing, and prices a unit of the stock by the loss times how much the unit raises the hazard. In climate economics this is the standard way to price tipping points (Cai and Lontzek, 2019; van der Ploeg and de Zeeuw, 2018; van den Bremer et al., 2026), and aggregate damage estimates take that price as an input (Dietz et al., 2021).

This paper shows that the hazard is not a sufficient primitive for that price. Three uncertainties surround a boundary: where it lies, when the random shocks will carry the system across it, and how fast the boundary moves toward the state as the budget is spent. A hazard summarizes the first two in one probability of crossing per unit of time. The price of restraint depends on the third. Taking the shock process as given, spending a unit of the budget does not change the shocks; it shortens the margin they have to cover, the distance between the state and the boundary, and so brings the crossing forward. This rests on one feature of the problem: the shocks can carry the system across the boundary before the budget is exhausted. Climate tipping elements are not only pushed over by the forcing; close enough to a threshold, random shocks, such as the year-to-year fluctuations of weather and ocean, can trigger the transition before the forcing does, which the literature calls noise-induced tipping (Ashwin et al., 2012). Stopping emissions therefore freezes the margin rather than removing the risk. Far from the threshold such a crossing is exponentially unlikely; what matters for the price is that it becomes likely close to it, where the shocks are as large as the margin. What a unit buys is delay, and its value turns on how fast the boundary approaches. If the boundary stands still, or approaches in proportion to the budget, the price stays bounded; if it approaches faster, as the square root of the budget left, the price rises steeply as the budget runs out. A hazard can be written to carry either rate, and nothing in the hazard decides which.

I make three claims. The first is that the hazard leaves this rate free. Its shape near the threshold, where the price is decided, cannot be checked against experience, since the system has never been there; it is a modelling choice. Two hazards that coincide on everything observed so far can therefore imply a price that stays bounded or one that diverges as the budget runs out. Even the whole path of the discounted probability of crossing does not fix the price: a fold and a pitchfork, two shapes of threshold whose discounted probability of crossing closes at the same rate along the path, carry prices of opposite classes, because spending the budget moves the fold's boundary and leaves the pitchfork's in place. In the cascade of van den Bremer et al. (2026), built on the

hazard of [Cai and Lontzek \(2019\)](#), the price of the first tipping point, at today's rate, sits at 2.49% of the highest value its own functional form allows, and two parameters the model takes from outside, the threshold temperature and the slope of the hazard above it, can move it to any level. The second claim is that the data cannot supply the rate in time. On the instrumental record of the Atlantic overturning circulation (AMOC), a sea-surface-temperature fingerprint ([Caesar et al., 2018](#)), even the most powerful test, at the central threshold and against the least favourable drift of the noise, detects an approaching boundary with even odds only once 58% of the remaining budget has been spent, and none of the tests I compute on that record tells the two classes apart. The third claim is that the dynamics of the system supply the rate. At a fold, where a stable and an unstable state approach and merge, the unstable state, i.e. the boundary, moves toward the stable one as the square root of the remaining budget, and the price inherits that rate. In physical models, the AMOC ([Cessi, 1994](#)), the West Antarctic ice sheet (WAIS) ([Schoof, 2007](#)) and the Amazon forest ([Staal et al., 2020](#)), three of the main climate tipping elements ([Lenton et al., 2008](#)), are folds.

What sets the price is whether the boundary stands still or moves toward the state as the budget is spent. A system rests in a stable state and tips if pushed past an unstable one; the gap between the two is the safety margin, and a budget can close that gap in two ways (Figure 1).

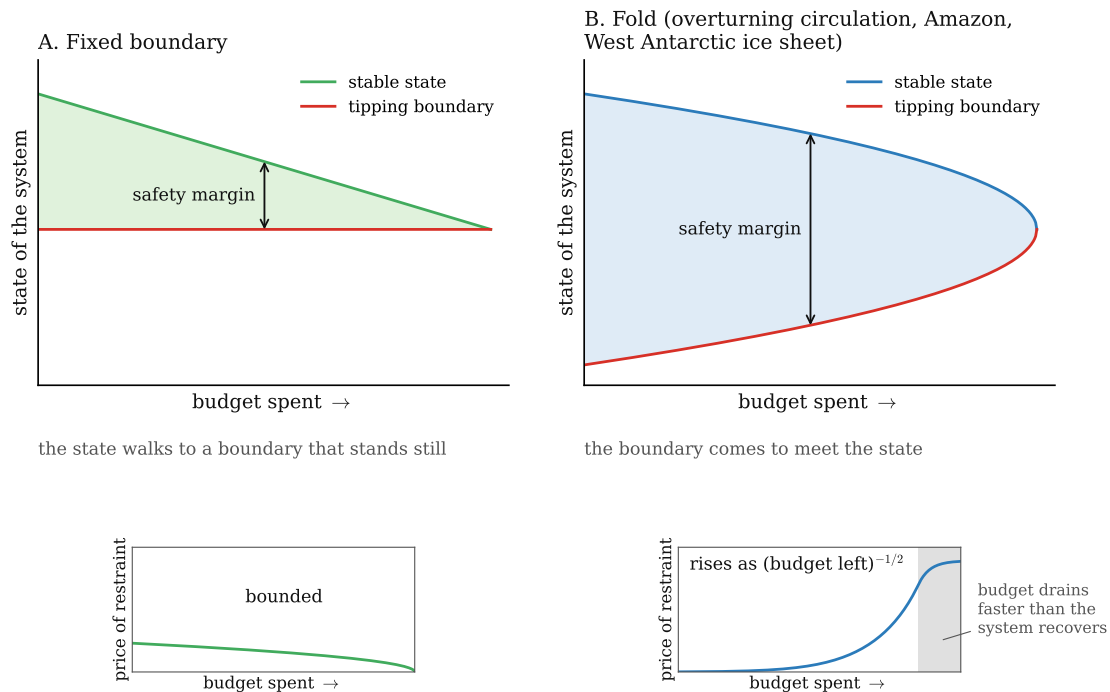


Figure 1: Whether the boundary moves. Left: boundary stands still; price of restraint stays bounded (hazard models in use). Right: boundary moves toward the state; price rises as the inverse square root of the budget left, until the budget drains faster than the system recovers (shaded).

In a structural model of default (Merton, 1974; Leland, 1994) the boundary stands still and the firm's value walks toward it: the margin closes linearly, and the claim that pays at default keeps a finite slope. At a fold the boundary itself moves: the unstable state comes toward the stable one as the budget is spent, and the two merge. Each unit spent then moves the boundary further than the last, and the marginal value of restraint rises as the inverse square root of the budget left, until the budget drains faster than the system recovers. Weitzman (2009) obtains an unbounded price from an unbounded loss and a fat tail; here the loss is finite, the shocks are Gaussian, and the steep rise comes from the geometry of the approach. The rise has an economic reading. The price is steep only where the shocks are as large as the margin, that is, where crossing within a discounting time has become likely; far from the threshold it is exponentially small. A price signal is therefore weak while acting is cheap, and strong once crossing is likely. In the hazard models of climate economics there is no boundary in the state: the distance to a threshold is not an object of the model, so nothing in it closes faster than the hazard is written to.

The damage fixes only the level and the sign of the price; the geometry fixes its shape. The paper's three folds play different roles accordingly. Where the argument is about the shape of the price or about what a record can detect, the damage cancels, and I use the AMOC: it alone has a long instrumental record (Caesar et al., 2018), on which its recovery rate can be measured (Boers, 2021), and its threshold is uncertain over a wide published range, from 1.4 to 8.0 °C (Armstrong McKay et al., 2022). Across that range its tipping price changes by a factor of 5.83. Replacing the fold by the class of the hazard models in use changes it by 2.65 at the near end of the range today, and by up to 6.1 as the budget is spent: by the end, the choice of class, which the literature makes implicitly, weighs as much as the whole disagreement over where the threshold lies, which it debates. Where the argument is about a date, the sign of the damage matters, and for the AMOC it is contested: Anthoff et al. (2016) find small net benefits from a shutdown, a result Keen et al. (2022) reject as at odds with projected losses in food production. The dates therefore rest on the Amazon and the WAIS, whose damages have an uncontested sign.

The geometry of the boundary changes what policy can rely on. In the leading synthesis of tipping points, Amazon dieback adds about 0.1% to the social cost of carbon (Dietz et al., 2021), an estimate that has been disputed (Keen et al., 2022) and that its authors call probably a lower bound (Dietz et al., 2022). That debate is about the size of the damage; this paper adds that near a fold the price is steep and that the hazard does not identify its shape. What policy can rely on without the damage is a date: the date after which stopping emissions can no longer avoid the threshold, the warming left before it divided by the pace at which it is spent. In the terms of Weitzman (1974), a quantity is the better instrument when the benefit of restraint curves more than its cost, as it does near a fold, and the date is that quantity: a cap on cumulative emissions, like the remaining carbon budgets for temperature targets, which complements any premium on the social

cost of carbon. It needs neither the damage, nor the noise, nor the exponent. For the Amazon it is 78 years on the high-emissions path SSP3-7.0 at the median, and under 37 in one draw in ten. These are latest dates, each of the following assumptions delaying them: emissions stop at once, CO₂ alone counts, warming is the only pressure on the forest, leaving out deforestation, and the Amazon does not interact with other elements. Whether avoiding the threshold is worth its cost needs the damage, which this paper does not estimate; until when it can still be avoided needs none, and the date answers it now, while a price signal would come too late.

This paper contributes first to debates on climate change, where it speaks to three literatures. The first prices climate tipping in integrated assessment models, either through a hazard (Cai and Lontzek, 2019; van der Ploeg and de Zeeuw, 2018; van den Bremer et al., 2026) or through a threshold at an unknown location that the planner learns about (Lemoine and Traeger, 2014; Fillon et al., 2023; Liski and Salanié, 2026). Both deliver a bounded premium: the first because of where its parameters sit, the second because uncertainty about where the threshold lies cannot by itself produce a divergence. I show that the hazard does not identify the object these models price, and that the dynamics of the element do. Folds are familiar from the economics of shallow lakes, where the same pollution can leave a lake clear or turbid (Mäler et al., 2003; Brock and Starrett, 2003), and Polasky et al. (2011) compare risks of such a shift that management can change with risks it cannot. The problem studied here is of the first kind. I add the random shocks that can push the system across, and ask what one unit of the budget is worth. The second detects the slowing of recovery as a fold nears (Scheffer et al., 2009; Boers, 2021) and extrapolates it to a date (Ditlevsen and Ditlevsen, 2023) that does not survive changes of data or of noise (Ben-Yami et al., 2024). Recent work asks what such warnings are worth for decisions (Diekert et al., 2025), and in ecology they can come too late when the driver of the shift can only be reduced slowly (Biggs et al., 2009). I bound what any test could detect before the budget is spent. The third is the choice of instrument (Weitzman, 1974; Hoel and Karp, 2001; Newell and Pizer, 2003). The date connects the pricing problem to that choice: near a fold the price cannot guide policy in time, and a quantity can.

More generally, the paper contributes to the pricing of risk near a boundary. Structural models of default price claims near a boundary that stands still, so that the margin closes linearly (Merton, 1974; Leland, 1994), and macro-finance studies how feedbacks amplify the approach to a crisis (Brunnermeier and Sannikov, 2014). The fold is the case in which the boundary itself moves toward the state, and I show that this changes the class of the price, not only its level.

Section 2 sets out the model and the price, and shows that the hazard leaves the exponent free and that the record cannot estimate it in time. Section 3 derives the exponent from the geometry of the boundary and assigns the climate tipping elements to their class. Section 4 measures what the choice of class moves, and turns the problem into deadlines that need no damage estimate.

2 The price of restraint and its missing exponent

This section builds the price of restraint and shows what it does not depend on. I set out the model, derive the price from the planner's problem as the response of the crossing value to the budget, and reduce the question of whether it diverges to a single exponent. I then show that the representations in use leave that exponent free, and that the instrumental record cannot supply it in time. What a tonne buys is delay, and the models in use do not say how much.

2.1 The model and the price

A state X_t evolves under a controllable drift against noise,

$$dX_t = f(X_t; \mu) dt + \sigma dW_t, \quad (1)$$

where f is the drift, the force that pulls the state back toward its resting point, σ the size of the noise, and W a Brownian motion. The budget is $\mu = M^* - M = M^* \epsilon \geq 0$: M is the stock the control accumulates, M^* the stock at which the threshold is reached, and ϵ the fraction of the budget still unspent, which I call the proximity. The budget enters through the drift. For each $\mu > 0$ the drift vanishes at two points, a stable rest point $x^*(\mu)$, where $f' < 0$ so that a displaced state returns, and an adjacent unstable point $x_-(\mu)$, where $f' > 0$ so that a displaced state runs away; the second is the boundary, and the state that reaches it does not come back. Lowering μ moves the two points toward each other. The gap between them is the safety margin $d(\mu) = x^*(\mu) - x_-(\mu)$, positive and closing to zero as $\mu \downarrow 0$. Crossing the boundary, at the first time $\tau = \inf\{t : X_t \leq x_-(\mu_t)\}$ at which the state reaches it, inflicts a one-time welfare loss $\mathcal{L}$, and the planner discounts at rate ρ .

The loss $\mathcal{L}$ is paid at the random time τ , so its value today is $\mathcal{L}$ times the expected discount factor at that time, $\mathbb{E}_x[e^{-\rho\tau}]$. Write $\varphi(x; \mu)$ for that expectation, starting from x with the budget at μ : it is the price today of a claim paying one unit on crossing, it equals one at the boundary, where crossing is immediate, and it falls toward zero as the state sits far from it. The value today of the loss is $\mathcal{L}\varphi$. The equation φ solves comes from a one-step argument. Hold the budget fixed and start the state at x . Over a short interval dt , either the state crosses and the claim pays, or it does not, in which case the planner holds the very same claim an instant later, started from the new position and discounted by one instant: $\varphi(x) = \mathbb{E}_x[e^{-\rho dt}\varphi(X_{dt})]$. Expanding both factors to first order in dt , with $\mathbb{E}[dX] = f dt$ and $\mathbb{E}[(dX)^2] = \sigma^2 dt$ read off (1) (Itô's lemma),

$$\varphi = (1 - \rho dt) \left(\varphi + f\varphi' dt + \frac{\sigma^2}{2}\varphi'' dt \right) + o(dt),$$

and cancelling φ and dividing by dt leaves

$$\underbrace{\frac{\sigma^2}{2} \varphi''}_{\text{diffusion}} + \underbrace{f \varphi'}_{\text{drift}} = \underbrace{\rho \varphi}_{\text{discounting}}, \quad x \in (x_-, \infty), \quad \varphi(x_-) = 1, \quad \varphi \text{ bounded as } x \rightarrow \infty. \quad (2)$$

The two boundary conditions are read off the definition: at x_- crossing is immediate, $\tau = 0$, so $\varphi(x_-) = 1$; far from the boundary crossing is delayed but not prevented, so φ stays bounded. The budget enters through the drift: spending it lowers μ , which brings the unstable state toward the stable one and shortens the distance φ has to discount over¹.

The price of restraint is the marginal cost, in welfare, of spending one more unit of the budget. To obtain it I place the crossing in a growth model. Capital K produces output RK , is consumed at the rate C , and pays for abatement $\Gamma(z)K$, the cost of abating a share $z \in [0, 1]$ of gross emissions E_0 , a fraction $\Gamma(z)$ of capital; so $\dot{K} = RK - C - \Gamma(z)K$ and $\dot{M} = (1 - z)E_0$. The planner maximizes the discounted logarithm of consumption, and crossing destroys a fraction D of capital. The value function $V(K, M, X)$ solves

$$\rho V = \max_{C, z \in [0, 1]} \left\{ \underbrace{\log C + V_K(RK - C - \Gamma(z)K)}_{\text{consumption and capital}} + \underbrace{V_M(1 - z)E_0}_{\text{carbon stock}} + \underbrace{f(X; M) V_X + \frac{\sigma^2}{2} V_{XX}}_{\text{climate state}} \right\}.$$

Three assumptions separate this problem. Utility is logarithmic, technology is linear in capital, and the loss is a fraction of capital. Under them the value function is additive, $V = \frac{1}{\rho} \log K + v(M, X)$, up to a constant that I absorb in v . Then $V_K = 1/(\rho K)$, and the first-order condition for consumption, $1/C = V_K$, gives $C = \rho K$. Substituting into the equation for V , the left-hand side is $\log K + \rho v$ and the right-hand side is

$$\log K + \log \rho + \frac{R - \rho}{\rho} + \max_{z \in [0, 1]} \left\{ -\frac{\Gamma(z)}{\rho} + v_M(1 - z)E_0 \right\} + f(X; M) v_X + \frac{\sigma^2}{2} v_{XX},$$

so $\log K$ cancels, and the constants $\log \rho + (R - \rho)/\rho$ are absorbed in v . The loss of a fraction D of capital lowers $\log K$ by $\log(1 - D)$, so it costs $\mathcal{L} = -\frac{1}{\rho} \log(1 - D)$ in welfare, whatever the level of capital. What remains is an equation for the climate part v alone,

$$\frac{\sigma^2}{2} v_{XX} + f(X; M) v_X - \rho v = - \max_{z \in [0, 1]} \left\{ -\frac{\Gamma(z)}{\rho} + v_M(1 - z)E_0 \right\}, \quad v = g(M) - \mathcal{L} \text{ at the boundary}, \quad (3)$$

where g is the part of v that depends on the stock alone. The right-hand side involves the climate state only through v_M , the response to the stock. Holding the stock fixed while the state crosses,

¹I compute φ holding the budget fixed while the state moves. Strictly, the budget falls meanwhile, and equation (2) gains the term $\dot{\mu} \partial \varphi / \partial \mu$ on its left-hand side. Dropping it is accurate when the budget drains slowly relative to the system's own recovery; Section 3.3.3 states where this holds, and Appendix A.3 computes the price without it.

as in (2), the part of v that depends on the state solves (2) multiplied by $-\mathcal{L}$, so it is $-\mathcal{L}\varphi$; the rest of v depends on the stock alone and carries the cost of abatement. Each assumption is needed: a subsistence floor would keep consumption from being a fixed share of wealth, a power utility would multiply the two blocks instead of adding them, and decreasing returns would leave K in the climate equation. This is the growth structure of [Pindyck and Wang \(2013\)](#) in its log case.

The carbon price therefore has two components. The Ramsey component is the part of the price that a standard integrated model computes from the stock alone. The threshold component $\text{SCC}_{\text{threshold}}$ prices the crossing, and is what this paper computes: it is $\mathcal{L} |\partial\varphi/\partial M|$ at the current state, which one more unit of emissions does not move, in welfare units; the factor ρK that converts welfare into consumption multiplies both components and reaches neither the exponent nor the sign of the noise response. Since $\mu = M^*\epsilon$, a unit of stock is a fraction $1/M^*$ of the budget, and $\partial/\partial M = -(1/M^*) \partial/\partial\epsilon$, so

$$\text{SCC}_{\text{threshold}} = \frac{\mathcal{L}}{M^*} \left| \frac{\partial\varphi}{\partial\epsilon} \right|. \quad (4)$$

The price therefore reduces to one question: how sensitive is the crossing value φ to the budget?

2.2 One exponent decides

The price (4) is the slope of the crossing value in the budget. The crossing value itself is bounded: it lies between zero and one, and it tends to one as the budget runs out, since with no margin left crossing is immediate. A bounded function can still have an unbounded slope, and whether it does depends on how fast it approaches one. Measure that speed by the shortfall $1 - \varphi$, the distance of the crossing value from certainty, and suppose first that it closes as a power of the remaining budget,

$$1 - \varphi \approx C_\varphi \epsilon^p \quad \text{near } \epsilon = 0, \quad \text{with } C_\varphi > 0 \text{ and } p > 0.$$

Differentiating gives

$$\left| \frac{\partial\varphi}{\partial\epsilon} \right| \approx C_\varphi p \epsilon^{p-1} = \frac{p(1 - \varphi)}{\epsilon}.$$

If $p < 1$ the exponent $p - 1$ is negative, and the slope, hence the price, grows without bound as the budget runs out: the shortfall is still large when little budget remains, so the last units close it steeply. If $p = 1$ the slope tends to the constant C_φ , and if $p > 1$ it tends to zero. When the shortfall is not exactly a power, p is its leading power,

$$p = \lim_{\epsilon \rightarrow 0} \frac{\log(1 - \varphi)}{\log \epsilon},$$

the slope of $\log(1 - \varphi)$ against $\log \epsilon$; slower factors such as logarithms do not change it. The exponent p is the object of the paper. Every representation of the crossing implies a value of it, and Section 2.3 asks what the representations in use imply.

In a model with a state, one more step is needed. The crossing value then depends on the state as well as on the budget, $\varphi(x; \epsilon)$, and a unit of emissions moves the budget but not the state. The price is therefore the slope in the budget at the current state, and it follows how fast the budget moves the boundary toward that state. Section 3 shows that at a fold this gives the exponent one half, and that a geometry whose boundary stands still gives a bounded price.

What a marginal unit of restraint buys is a later crossing, not an avoided one. Because the noise can carry the state across from any margin, stopping emissions freezes the margin rather than removing the risk: with a frozen margin the expected wait before crossing grows exponentially with the height of the barrier the noise must climb, but it is finite. The price of restraint is therefore the value of the time a unit adds before the loss. Under a hazard that is zero below a floor, as in Cai and Lontzek (2019), holding the stock below the floor removes the risk instead, and restraint buys an outcome. The crossing I price combines the two mechanisms that Ashwin et al. (2012) distinguish: the noise carries the state across, and spending the budget brings the bifurcation closer, which lowers that barrier.

2.3 What the models in use leave open

The economics of climate tipping represents a threshold in two ways: as a hazard, a probability of crossing per unit of time that rises with warming or with the carbon stock, or as a hard threshold at a location the planner does not know and learns about. I ask what exponent each implies. The hazard leaves it free, the values in use make the price bounded, and a threshold of unknown location cannot make it diverge.

First, for the hazard, Cai and Lontzek (2019) make it linear in temperature above a floor, and van der Ploeg and de Zeeuw (2019) linear or quadratic in the carbon stock. With a constant rate λ , crossing has not happened at time t with probability $e^{-\lambda t}$, so the crossing value of Section 2.1 is

$$\varphi = \mathbb{E}[e^{-\rho\tau}] = \int_0^\infty e^{-\rho t} \lambda e^{-\lambda t} dt = \frac{\lambda}{\rho + \lambda}.$$

The rate is written as a function of a climate state x , temperature or the carbon stock, which itself depends on the stock M . Holding the rate at its current value², the price is the loss times the

²Along the path the rate rises anyway. For a given loss, holding it fixed therefore overstates the price whenever the increase in the rate that a tonne causes does not grow along the path, as when the rate is linear in a state that is itself linear in the stock, which is the case of Cai and Lontzek (2019) and van den Bremer et al. (2026); any upper bound on this price, such as the one derived below, then also bounds the price along the path.

derivative of φ in M :

$$\text{SCC}_{\text{threshold}} = \frac{\mathcal{L} \rho}{(\rho + \lambda)^2} \underbrace{\lambda'(x)}_{\text{hazard in the state}} \underbrace{x'(M)}_{\text{state in the stock}}. \quad (5)$$

Near the threshold the rate is large against the discount rate, so $\rho/(\rho + \lambda)^2 \approx \rho/\lambda^2$, and since $1/\lambda$ is the mean wait before crossing, (5) reads

$$\text{SCC}_{\text{threshold}} \longrightarrow \mathcal{L} \rho \frac{1}{\lambda^2} \frac{d\lambda}{dM} = \mathcal{L} \rho \left| \frac{d}{dM} \frac{1}{\lambda} \right| : \quad (6)$$

the loss, times the discount rate, times the years of waiting that one more tonne removes.

The price therefore diverges only if the mean wait shrinks infinitely fast as the last units of the budget are spent, and nothing in the hazard says whether it does. Write $\lambda = c_0 \epsilon^{-p}$, for a constant $c_0 > 0$: the mean wait is ϵ^p/c_0 , and by (6) the price goes as ϵ^{p-1} . Every $p > 0$ gives a smooth rate, and the formalism does not say which to write. At the values in use, the rate and its slope stay finite up to the threshold, the mean wait shrinks at most linearly, and the price is bounded. Observations cannot decide either: two hazards can agree at every state observed so far and differ near the threshold, where the system has never been.

In their model, [van den Bremer et al. \(2026\)](#) solve a cascade of tipping points with the hazard of [Cai and Lontzek \(2019\)](#): the first tipping point arrives at the rate $\lambda_1 = h_1(T - T^*)$, zero below a threshold temperature T^* and rising with current warming T above it at the slope h_1 ; the later ones follow at constant rates, so they multiply the loss and add nothing to the sensitivity of the price to carbon. Temperature rises with the stock at the rate $\chi = dT/dM$, so $d\lambda_1/dM = h_1\chi$, and with $\Lambda = \lambda_1/\rho$ the rate in units of the discount rate, (5) becomes

$$\text{SCC}_{\text{threshold}} = \frac{\mathcal{L}\chi}{T - T^*} \cdot \frac{\Lambda}{(1 + \Lambda)^2}, \quad \sup_{\Lambda} \text{SCC}_{\text{threshold}} = \frac{\mathcal{L}\chi}{4(T - T^*)}. \quad (7)$$

The ceiling does not depend on the slope and grows without bound as the threshold approaches current warming, so the model's own form spans prices from negligible to unbounded. Two parameters it imports decide where the price falls: the threshold T^* , uncertain by degrees in the physical literature, and the slope, set at $h_1 = 0.35\% \text{ } ^\circ\text{C}^{-1} \text{ yr}^{-1}$, a value [Cai and Lontzek \(2019\)](#) calibrate on the expert survey of [Kriegler et al. \(2009\)](#). At the published calibration the threshold sits at $T^* = 1 \text{ } ^\circ\text{C}$, just below the model's current warming of $1.1 \text{ } ^\circ\text{C}$, so the rate today is 174 times below the one that would maximize the price. Evaluated at today's rate, the tipping price is 2.49% of the ceiling, and raising the slope alone multiplies it by 40.1³.

³The published loss also grows with warming, $\mathcal{L} \propto T^\nu$. The price (7) then gains the factor $1 + \nu s(1 + \Lambda)$, with $s = (T - T^*)/T$ the share of current warming above the threshold, and its ceiling the factor $(1 + \nu s)^2$, which stays

The second representation does not supply the exponent either, and it cannot produce a divergence (Lemoine and Traeger, 2014; Fillon et al., 2023; Liski and Salanié, 2026). Let the threshold lie somewhere below a stock b , the top of the planner’s prior, and let emissions run at the rate E_0 . Crossing happens at the latest when the stock reaches b , so $\tau \leq (b - M)/E_0$, and

$$1 - \varphi(M) = \mathbb{E}[1 - e^{-\rho\tau}] \leq \rho \mathbb{E}[\tau] \leq \rho \frac{b - M}{E_0}.$$

The shortfall closes at most linearly in the remaining stock, so $p \geq 1$ and the price stays bounded, for every prior on a bounded range. The implied hazard does run away at the top of the prior, but the mean wait, bounded by $(b - M)/E_0$, cannot shrink infinitely fast, as (6) would need. For a prior whose density behaves near the top of its support as a power of the distance to it, which covers the usual priors, conditioning on survival gives $p = 1$ exactly.

Which exponent to write is therefore the question the representations leave open; Section 3 answers it from the dynamics of the element.

2.4 What the record can establish

A hazard observed far from the threshold says nothing about its exponent. A model with a state does better: how fast the state recovers from its fluctuations carries information about how the boundary moves, and one might hope to read the exponent from the observed historical record. But it cannot be read in time; Appendix B gives the details. An approaching fold shows in data chiefly as a slowing down: the element returns to rest more slowly after each fluctuation, because the force that pulls it back weakens as the two states draw together. Two facts make this signal late. The slowing is gradual, so what a record can show of it is set by the length of the record, not by how close the threshold is. For the trend statistic used in this literature to see a fold with even odds even at the threshold, the record would have to be some 349 years long. And the record lengthens only as time passes, which at constant emissions means as the budget is spent. Detection and spending run on the same clock: a record long enough to see the approach is one over which much of the budget is already gone.

How late can be computed by comparing two worlds. In the first the element approaches a fold, and its recovery rate falls as the square root of the time left before the threshold, which at constant emissions falls by one year every year. In the second it does not: the recovery rate stays constant, and the size of the fluctuations drifts in the way that makes the two worlds hardest to tell apart. The record cannot exclude such a drift. I estimate the trends in the recovery rate and in the noise jointly, under 120 specifications that vary the sample period, the detrending method, the

bounded, so the ceiling is still unbounded in T^* . The share and the factor above use this loss.

block length of the bootstrap and the estimator. In 120 of them, the 95% confidence region of the two trends contains both a trend in the recovery rate alone and a trend in the noise alone: the record is consistent with either explanation. A test applied to the record sometimes raises a false alarm in the second world and sometimes detects the approach in the first; the probability of detecting it is the power of the test. Holding false alarms at one in twenty, I compute the highest power that any test can reach, that of the likelihood-ratio test: within Gaussian models of the record, no test detects the approach more often. Figure 2 computes this bound on the 146 annual means of a sea-surface-temperature fingerprint of the AMOC (Caesar et al., 2018), the one tipping element with a long instrumental record.

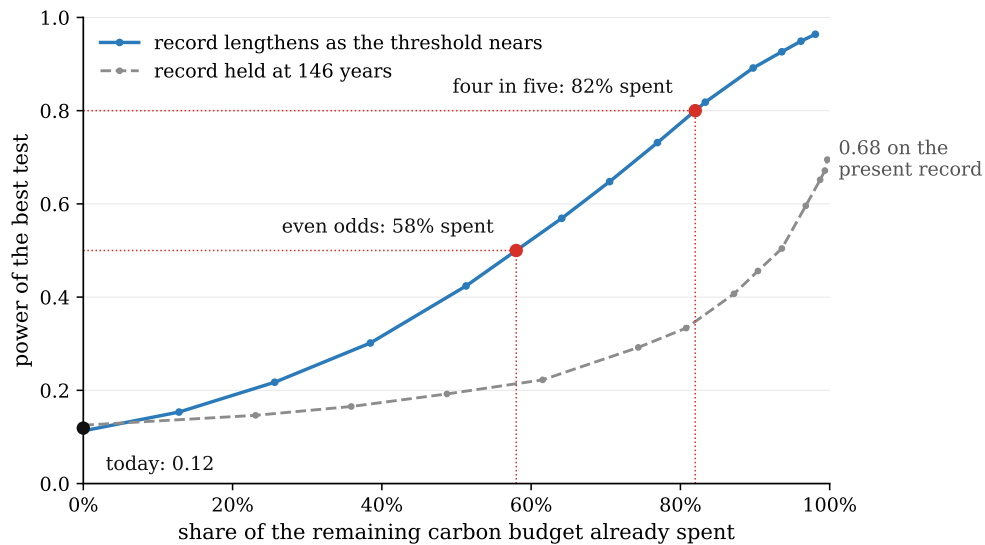


Figure 2: What the record can establish: the power of the best test on the instrumental record of the overturning circulation, against the share of the remaining budget spent. Solid: the record lengthens as the threshold nears. Dashed: the record keeps its present length.

To draw Figure 2 I simulate the record as annual means of a continuous process whose recovery rate reproduces the lag-one autocorrelation of the detrended series. At central inputs the threshold lies 156 years away at constant emissions. Moving along the horizontal axis, the budget is spent at today's rate: each year adds one annual mean to the record and removes one year from the time left. At each point I compute the bound by Monte Carlo, holding false alarms at one in twenty. The solid line lets the record lengthen in this way; the dashed line keeps it at its present length. Today the bound is 0.12. If the record lengthens as the threshold nears, even the best test detects the approach with even odds only once 58% of the remaining budget has been spent, and with odds of four in five only after 82%. If the record keeps its present length, the bound never exceeds 0.68. Nor can the record tell the two classes apart: the power is the same, to within 0.04, whichever class the test assumes. By the time the record could settle the question, most of what the answer would

govern is gone. The record has been used to test for an approaching collapse of the circulation and to date it (Ditlevsen and Ditlevsen, 2023), and such dates have been shown to depend on the data and assumptions used (Ben-Yami et al., 2024).

The record, then, cannot tell the planner how the boundary moves before most of the budget is spent. The dynamics of the element can: the physical literature reports how each element loses stability, from process models and palaeorecords, and Section 3 shows that this, with the rate at which the budget moves the element’s parameter, fixes the exponent.

3 Geometry fixes the exponent

Section 2 left one number open: the exponent of the price. The models in use do not fix it, and the record cannot estimate it in time. This section shows that the dynamics of the element fix it, and this is the main contribution of the paper. The dynamics matter because they say how the budget moves the boundary. When the boundary stands still and the state walks toward it, the price stays bounded, whatever the rate at which the margin closes. When the budget moves the boundary itself toward the state, as at a fold, where a stable and an unstable state approach and merge, the boundary moves as the square root of the budget, and the price rises as the inverse square root of the budget left, until the budget drains faster than the system recovers.

The argument is about the fold, the geometry the physical literature reports for the AMOC, the Amazon and the ice sheets, and the fold is what the policy section prices. I carry two other geometries through the construction, the transcritical and the pitchfork, whose boundaries stand still, so that what decides the class is visible: how the budget moves the boundary⁴. Section 3.1 sets out the three geometries, the climate elements the physical literature assigns to them, and the noise layer, the unit in which distance to the threshold is measured; Section 3.2 states the three properties of the dynamics the result needs; and Section 3.3 derives the price step by step.

3.1 Three geometries, one contrast

Take the fold first. Its drift is $f(x; \mu) = \mu - x^2$, so the state of equation (1) moves as $dX = (\mu - X^2) dt + \sigma dW$, and the crossing value solves (2) with this drift, $\frac{\sigma^2}{2} \varphi'' + (\mu - x^2) \varphi' = \rho \varphi$. The two rest points, where the drift vanishes, sit at $\pm\sqrt{\mu}$, and they merge at the origin when the budget is gone. The square root does not depend on the particular drift $\mu - x^2$. In general, when a

⁴The physical literature describes other ways of losing stability, which I leave for extensions. An element can lose its resting state to an oscillation (a Hopf bifurcation), as the El Niño–Southern Oscillation is often modelled (Jin, 1997); whether its price stays bounded then depends on whether the size at which the oscillation does damage moves with the budget, which this paper does not model. Tipping caused by the pace of the forcing, and transitions that spread through space, are discussed in the conclusion.

stable and an unstable state merge as one parameter moves, the drift and its slope vanish together at the merging point; if the curvature of the drift and its response to the parameter do not, the margin closes as the square root of the distance to that point.

Figure 3 shows three shapes a threshold can take, drawn as a potential U , with $f = -\partial U/\partial x$: the state rests at the bottom of a valley and the boundary sits on a ridge. While the budget lasts the two are apart; spending the budget brings them together, and what separates the geometries is whether it does so by moving the boundary.

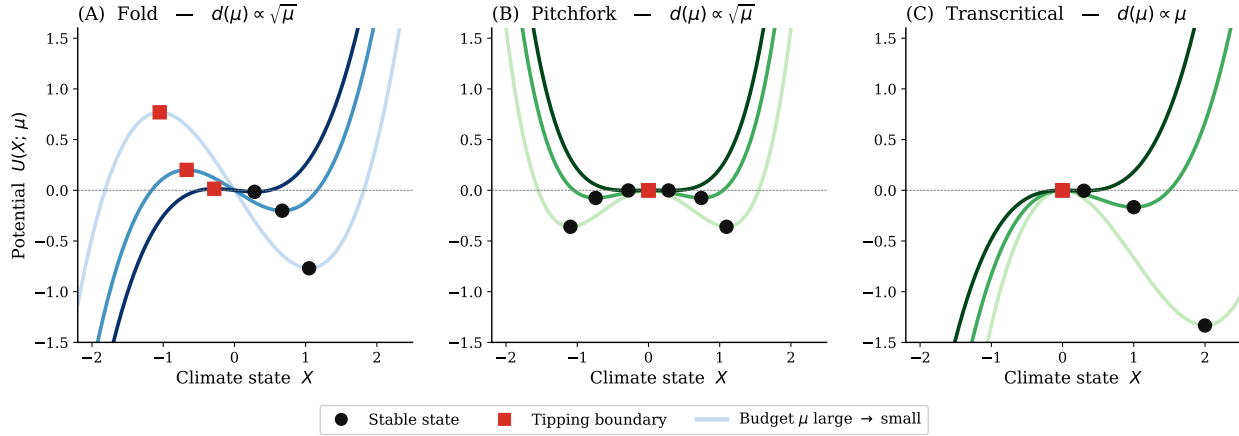


Figure 3: The three geometries of Table 1 as the budget falls, drawn as potentials. Blue: the fold, whose boundary moves. Green: the transcritical and the pitchfork, whose boundary stands still. Darker curves: smaller budgets.

The first contrast is between the fold and the transcritical, one dynamical system written twice. In the coordinate $y = x - \mu/2$ the transcritical drift $\mu x - x^2$ becomes $\mu^2/4 - y^2$, the fold with the parameter $\mu^2/4$ in place of μ . The two share every local feature at the merged point and differ only in how the budget enters the parameter: in proportion in the fold, as its square in the transcritical. In the coordinate x , which the emissions leave in place, the transcritical's boundary stays at the origin while the stable state walks toward it, so the two geometries fall in opposite classes⁵.

Another geometry shows that the type of bifurcation and the closing rate of the margin do not decide the class either. The pitchfork $f = \mu x - x^3$ closes its margin at the fold's rate: the margin is $\sqrt{\mu}$, and along the path its crossing value follows the fold's power law. But its boundary is the unstable state at the origin, which does not move; it is the stable states that walk toward it. Its price at a fixed state therefore falls to zero as the budget runs out, as the transcritical's does. Thus, two

⁵Relabelling the state by any increasing function that does not depend on the budget changes neither the crossing time nor the price, so the class does not depend on whether an element is measured in degrees or in a vegetation index. A relabelling that depends on the budget, such as $y = x - \mu/2$, is different: holding y fixed while a unit is spent moves the physical state x , which emissions do not, and in y the transcritical looks like a fold. The price holds fixed the state that emissions leave in place, which is x .

geometries whose crossing values close at the same rate along the path carry prices of opposite classes, because a unit of budget moves the fold's boundary and leaves the pitchfork's in place.

To say how close the state is to the threshold, the geometries need a unit of distance. At the merged point of the fold the restoring force vanishes, so there is a zone around it in which the noise, not the drift, moves the state. Its width follows from comparing the terms of (2). Suppose φ changes by an amount of order φ over a distance ℓ around the merged point. Then φ' is of order φ/ℓ and φ'' of order φ/ℓ^2 , so the diffusion term $\frac{\sigma^2}{2}\varphi''$ is of order $\sigma^2\varphi/\ell^2$. Over that zone the drift $-x^2$ is of order ℓ^2 , so the drift term is of order $\ell^2 \cdot \varphi/\ell = \ell\varphi$. Diffusion dominates on short distances and drift on long ones, and the two are of the same size when $\sigma^2/\ell^2 = \ell$, that is $\ell^3 = \sigma^2$, which gives $\ell = \sigma^{2/3}$. The discount term $\rho\varphi$ carries no power of ℓ , so in this unit it enters only through the ratio ρ/ℓ . I call ℓ the noise layer. It is the natural unit of distance to the threshold: measured in layer widths, the half-margin is

$$\eta^* = \frac{d(\mu)}{2\ell(\sigma)}, \quad (8)$$

and it decides whether the crossing value is close to one, inside the layer, or exponentially small, outside it.

The physical literature reports which of these geometries an element follows. It answers from process models and palaeorecords, without reference to what crossing would cost (Lenton et al., 2008; Armstrong McKay et al., 2022; Ashwin et al., 2012). What it does not report is the damage, the question climate economics has made its main one: the order of availability is the reverse of the order of attention. Read against the result of this section, the physical literature assigns each element to a class without any economic input. Table 1 collects the three geometries, how the budget moves the boundary, the order of the drift at the merged point, the width of the noise layer, and the elements assigned to each. The main climate tipping elements are folds. The transcritical and the pitchfork appear in idealized versions of the same elements: the Amazon forest in a vegetation model without the forest's feedback on rainfall (Cox, 2001), and the overturning circulation in a basin symmetric about the equator (Dijkstra and Molemaker, 1997). What separates them from a fold is reported: a transcritical has a single stable state at each budget, whereas the bistability and hysteresis found for these elements require a fold; and a pitchfork requires a symmetry between the hemispheres that the real Atlantic lacks, so that in an asymmetric basin one of its branches appears through a fold.

Table 1: The three geometries, their classes and the climate elements assigned to them

Geometry	Drift $f(x; \mu)$	$d(\mu)$	Boundary moves as	k	$\ell(\sigma)$	Pricing implication	Climate elements
Fold	$\mu - x^2$	$2\sqrt{\mu}$	$\sqrt{\mu}$	2	$\sigma^{2/3}$	Rises as $\epsilon^{-1/2}$ inside the noise layer, until ϵ_{dyn} ; finite deadline	AMOC, Greenland, WAIS, Amazon
Transcritical	$\mu x - x^2$	μ	fixed	2	$\sigma^{2/3}$	Bounded; the boundary does not move	Amazon, vegetation model without rainfall feedback
Pitchfork	$\mu x - x^3$	$\sqrt{\mu}$	fixed	3	$\sigma^{1/2}$	Bounded; the boundary does not move	AMOC, basin symmetric about the equator

k is the order of the drift at the merged point. Fold elements: the AMOC, from bistability in thermohaline box models (Cessi, 1994); Greenland, from multistability and hysteresis in ice-sheet models (Robinson et al., 2012); the WAIS, from the marine ice-sheet instability (Schoof, 2007); the Amazon, from moisture-recycling bistability (Staal et al., 2020). Other elements are left unclassified: e.g. Armstrong McKay et al. (2022) treat boreal permafrost as a strong but non-runaway feedback with limited large-scale bistability. Climate economics prices all these elements in the bounded class.

3.2 Three hypotheses

The fold is one example. The result extends to any element with three properties: spending the budget moves the boundary toward the state at a definite rate (H1); near the threshold the force that pulls the state back weakens gradually, as a power of the distance to the threshold (H2); and the delay that a unit of restraint buys is worth something to a planner who discounts, and worth more the more impatient the planner is (H3). I take them in turn.

First, the boundary moves at a definite rate (H1). As the budget runs out, the boundary approaches the point where it meets the stable state, x_c , as a power of the budget: $|x_-(\mu) - x_c| \sim c \mu^p$, with $c, p > 0$. Its speed follows the same power, $|x'_-(\mu)| \sim c p \mu^{p-1}$, as it does whenever the boundary is a simple root of the drift, which moves smoothly with the budget. For the fold, the boundary sits at $x_- = -\sqrt{\mu}$, so $c = 1$ and $p = \frac{1}{2}$. The speed matters, and not only the position, because the price is a derivative at a fixed state: it measures how far one more unit of the budget moves the boundary toward a state that the unit leaves in place. A geometry whose boundary does not move, as the transcritical and the pitchfork, has no such rate, and its price stays bounded. This p is the exponent of Section 2.2: in both, the price goes as ϵ^{p-1} .

Second, the force that pulls the state back weakens as a power of the distance (H2). When the budget is gone, the stable state and the boundary have met at one point, and nothing pulls the state back there: the drift is zero at that point, and so is its slope. Near the point, the drift is therefore small, and it grows only as a power of the distance from it. Measuring the state as its distance x

from that point, the assumption is that this power is finite,

$$f(x; 0) = -c_k x^k (1 + O(x)), \quad x \rightarrow 0, \quad (9)$$

with $c_k > 0$ a constant and $k \geq 2$ the power. For the fold, the drift with the budget gone is $-x^2$, so $k = 2$ and $c_2 = 1$. This is what lets noise and impatience enter as one number. Measure the distance from the merged point in units of a length ℓ , $x = \ell\eta$, in equation (2) for the crossing value there. Each derivative divides by ℓ and the drift $-c_k x^k$ becomes $-c_k \ell^k \eta^k$, so the equation reads $(\sigma^2/2\ell^2) \varphi_{\eta\eta} - c_k \ell^{k-1} \eta^k \varphi_\eta = \rho \varphi$. Dividing by ℓ^{k-1} and choosing $\ell^{k+1} = \sigma^2$ leaves

$$\frac{1}{2} \varphi_{\eta\eta} - c_k \eta^k \varphi_\eta = \tilde{\rho} \varphi, \quad \ell(\sigma) = \sigma^I, \quad m(\sigma) = \ell(\sigma)^{k-1} = \sigma^J, \quad \tilde{\rho} = \frac{\rho}{m(\sigma)}, \quad (10)$$

with $I = 2/(k+1)$ and $J = 2(k-1)/(k+1)$, both positive for $k \geq 2$. The length ℓ is the noise layer of Section 3.1, $\sigma^{2/3}$ for the fold. The rate m is the pace at which the noise moves the state inside the layer, and impatience matters only relative to it, through $\tilde{\rho}$, the only parameter left in (10). How steeply the crossing value falls near the threshold then depends on the noise and the discount rate alone, not on the budget, and a unit of the budget acts on the price only by moving the boundary.

Third, the delay a unit buys has a positive value that rises with impatience (H3). Call $\Phi(\tilde{\rho})$ the amplitude: how fast the crossing value falls for each layer width the state stands from the threshold. It is zero when $\tilde{\rho} = 0$, since a planner who does not discount gains nothing from delay, and it rises with $\tilde{\rho}$, since a more impatient planner values the same delay more; Appendix A.1 shows both, for every power k . That Φ is positive keeps the main term of the price from vanishing, so the price goes as ϵ^{p-1} and not as a higher power. That it rises with impatience fixes the direction in which the price responds to noise, as Section 3.3 shows.

3.3 The dichotomy

The result holds on what I call the fixed-budget regime. There the budget drains slowly compared with how fast the state recovers from a shock, so it can be held fixed while the state moves, and the state already sits inside the noise layer, where crossing has become likely. On that regime, under (H1)–(H3), the price satisfies, as ϵ decreases through it,

$$\text{SCC}_{\text{threshold}}(\epsilon, \sigma, \rho) = \mathcal{L} \underbrace{c p (M^*)^{p-1}}_{\text{geometry}} \underbrace{\frac{\Phi(\tilde{\rho})}{\ell(\sigma)}}_{\text{steepness}} \underbrace{\epsilon^{p-1}}_{\text{proximity}} (1 + o(1)), \quad (11)$$

so the price follows the power law ϵ^{p-1} : it rises as the budget runs down when $p < 1$, and stays bounded when $p \geq 1$ or when the boundary does not move. I call the right-hand side of (11) without the factor $1 + o(1)$ the leading law. The factor $1 + o(1)$ is close to one when the state sits well inside the noise layer and the budget drains slowly.

3.3.1 Derivation

The derivation takes three steps, one per factor.

First, near the boundary the shortfall is the steepness times the distance to it. The crossing value equals one at the boundary x_- . When the distance $x - x_-(\mu)$ is small against the noise layer, the scale on which φ varies, a first-order expansion from the boundary suffices⁶:

$$1 - \varphi(x; \mu) = \kappa(\sigma, \rho) (x - x_-(\mu)) + o(x - x_-(\mu)), \quad \kappa \equiv -\varphi'(x_-) > 0. \quad (12)$$

The steepness κ is the rate at which the crossing value falls away from the boundary, positive because φ falls. At the stable state the distance is the margin, so the shortfall there is $\kappa d(\mu)$.

Second, the steepness depends on noise and impatience only. Measure the distance in layer widths, $x = \ell\eta$. Close to the threshold, the equation for the crossing value is then the one at the merged point, which by (H2) contains $\tilde{\rho}$ and nothing else. How fast the crossing value falls per layer width therefore depends on $\tilde{\rho}$ alone: it is the amplitude $\Phi(\tilde{\rho})$ of (H3). Converting layer widths back into distance divides by ℓ , so $\kappa = \Phi(\tilde{\rho})/\ell(\sigma)$ ⁷. By (H3), $\Phi > 0$, so $\kappa > 0$ and the leading term of (12) does not vanish. At that order the steepness carries neither μ nor ϵ .

Third, the budget enters through the boundary alone. Spending a unit moves the stock, not the state, so the price is the derivative of (12) in μ at a fixed state x . Because κ does not depend on the budget, this derivative moves only the boundary: $|\partial\varphi/\partial\mu| = \kappa |x'_-(\mu)|(1 + o(1))$, and (H1) gives $\kappa c p \mu^{p-1} (1 + o(1))$. The stock and the budget move in opposite directions, $\mu = M^* - M$, so $|\partial\varphi/\partial M| = |\partial\varphi/\partial\mu|$; and with $\mu = M^*\epsilon$, $\mu^{p-1} = (M^*)^{p-1}\epsilon^{p-1}$. Multiplying by $\mathcal{L}$ gives (11). Where the boundary does not move, as in the transcritical and the pitchfork, the leading term vanishes and the price is bounded.

I check (11) on the numerical solution of (2) (Appendix A.2). The price is computed as the slope of the crossing value in the budget at a fixed state, over a grid of noise levels and discount rates. For every pair of the grid its local slope in $\log \epsilon$ approaches $-\frac{1}{2}$ as the budget falls, and at the smallest budget computed the slope farthest from it is -0.492 . For the transcritical and the

⁶The expansion, and its derivative in the budget below, need φ smooth in the state and in the budget, which holds because (2) is linear with coefficients smooth in both.

⁷For the fold the midpoint of the two equilibria is the merged point, and the two sit symmetrically about it; for the transcritical and the pitchfork the merged point is the boundary itself. Reading κ at the boundary, at the midpoint or at the merged point changes it only at the next order in η^* , so the three readings agree as $\eta^* \rightarrow 0$.

pitchfork the price at a fixed state falls to zero as the budget runs out, with local slopes of at most 1.00 and 0.50. The law therefore gets the exponent right. Its level is right only when the state sits deep inside the noise layer: at a half-margin of 0.1 layer widths the exact price is 1.1 times the law, and at 0.3 it is 1.9 times; these ratios barely change with the discount rate.

3.3.2 What each factor says

Each factor of (11) has a reading. The geometry factor $c p (M^*)^{p-1}$ says how the budget moves the boundary; it carries the exponent, and with it the class. The steepness $\Phi(\tilde{\rho})/\ell(\sigma)$ says how fast the crossing value falls with distance; it is set by the noise and the discount rate, and it moves the level of the price, never its shape. The proximity factor ϵ^{p-1} is the shape: for the fold, with $p = \frac{1}{2}$, the price grows as $\epsilon^{-1/2}$.

Read back into the representation of Section 2.3, the fold's crossing value can be read as an arrival rate. The shortfall is κd . Inverting the shortfall of a constant hazard, $1 - \varphi \approx \rho/\lambda$, the rate that reproduces it is $\rho/(\kappa d)$, so the fold behaves as the hazard

$$h(\epsilon) = \frac{\rho}{\kappa d(\mu)} = \frac{\rho}{2\kappa (M^*)^{1/2}} \epsilon^{-1/2},$$

the family $\lambda = c_0 \epsilon^{-p}$ at $p = \frac{1}{2}$. An arrival-rate model that wrote this rate would return a divergent price with the fold's exponent. Its level would be twice the fold's, because a hazard has no state and its price follows the whole shortfall, while the fold's price follows only the boundary's share of it. The pitchfork shows the other side: a hazard fitted to its crossing value, which also closes as $\sqrt{\epsilon}$, would price it as divergent, yet its boundary stands still and its price is bounded. Even the full path of the crossing value does not fix the price; the geometry does.

Noise enters the price only through the steepness $\kappa = \Phi(\tilde{\rho})/\ell(\sigma)$. More noise widens the layer over which the crossing value stays close to one, so the crossing value responds less to the margin. With $\ell = \sigma^I$ and $\tilde{\rho} = \rho \sigma^{-J}$ as in (10),

$$\frac{\partial \kappa}{\partial \sigma} = -\frac{1}{\sigma \ell(\sigma)} \left[I \Phi(\tilde{\rho}) + J \tilde{\rho} \Phi'(\tilde{\rho}) \right] < 0, \quad (13)$$

the bracket being positive because I and J are, and Φ and Φ' by (H3). At a given proximity the price of restraint therefore falls as the noise rises: when chance does more of the work, each unit of restraint buys less delay. This holds deep inside the regime for every geometry whose boundary moves and that satisfies (H2). It runs against the presumption that more uncertainty calls for more caution, but it concerns the marginal price, not the total risk: at a given proximity more noise raises the expected loss, crossing becoming both likelier and sooner, while each extra unit of restraint buys less.

3.3.3 Where the result holds

The fixed-budget regime is bounded on both sides. Near the threshold, it ends at the dynamic scale ϵ_{dyn} , the proximity at which the budget, in relative terms, drains as fast as the state recovers from a perturbation. Above that scale, freezing the budget costs little: for AMOC at its calibrated values, the frozen and the draining prices differ by at most 11% at today's proximity, whether the state sits inside the noise layer or at its edge, so what limits the approximation is the element's recovery, not impatience. Closer than that, the boundary moves while the state approaches it, and the price leaves the power law: computed with the budget draining (Appendix A.3), it first rises faster than the fixed-budget price, by a factor of up to 1.8, then flattens. Farther from the threshold, the regime ends where the stable state leaves the noise layer. Beyond that point the shocks are small against the margin, crossing is exponentially unlikely, and the price is exponentially small.

I call the window the proximities between ϵ_{dyn} and today's proximity ϵ_{now} : the part of the approach that lies ahead before the power law stops. The three elements I calibrate all sit outside the noise layer today, so the regime covers only the lower part of their window, and none of them is in it yet. When one will enter it is not identified: it depends on the physical distance between the element's two states, which no dataset reports. The result is therefore about the class of the element, which the physical literature reports, and about what happens once the element enters the regime, which it can do before its deadline: the price then rises steeply, and late. The deadline of the next section does not depend on the regime.

For the AMOC, the calibration places ϵ_{dyn} at the proximity 0.027. Between today's proximity and that point, the leading law multiplies the price by about 5.1; the law is accurate deep inside the noise layer and less so near its edge. This is the most the power law can add over the window, and it is the actual rise only if the circulation already sits inside the noise layer today. If it enters the layer later, the power law applies only to the end of the window and adds less; before that, the price is small and rises outside the law.

4 Policy

This section turns the result into policy. It measures which of the field's disagreements moves the price most. It then replaces the price near a fold, steep and unidentified, by a date: the date after which stopping emissions can no longer avoid the threshold. That date needs no damage estimate and can be ranked before the damage is known. Whether stopping by then is worth its cost depends on the damage, and on how far the price rises before the budget outruns the element's recovery. The classification the price rests on is available before the damage debate is settled, because whether the budget moves the boundary, as at a fold, is reported by the physical literature

and does not require a damage function (Section 3.1).

4.1 Which disagreement moves the price

The leading law (11) makes the price a power of the budget left, $\text{SCC}_{\text{threshold}} \propto \mu^{p-1}$ with $\mu = M^* \epsilon$, so beyond its level there are two ways to move it: move the threshold, which moves μ , or move p . The first is where the field's effort has gone and the second is where it has not. Differentiating,

$$\frac{\partial \log \text{SCC}_{\text{threshold}}}{\partial \log \mu} = p - 1, \quad (14)$$

a constant: the response to the budget left does not depend on how much is left, so approaching a threshold does not make its location matter more than it did. Replacing the exponent, by contrast, multiplies the price by a factor that grows as the budget is spent, until ϵ_{dyn} . The nearer an element sits to its edge, the more of its price turns on the exponent nobody estimates.

The comparison uses ratios of the price law to itself, so the level of the price and the damage cancel. It uses the leading laws, which describe the price once the state is inside the noise layer; the AMOC is outside that regime today and can enter it before its deadline (Appendix C.1). Its threshold ΔT^* , the warming above pre-industrial at which it tips, is reported at 4°C , with a published range of $[\Delta T_{\text{low}}^*, \Delta T_{\text{high}}^*] = [1.4, 8.0]$; current warming is ΔT_0 , so that the proximity is $\epsilon = (\Delta T^* - \Delta T_0)/\Delta T^*$, the warming left over the warming to the threshold. I change one assumption at a time. First, keep the fold and move the threshold across the published range: the budget to the threshold moves with it, and the price changes by a factor of 5.83 from one end of the range to the other. Second, keep the threshold and change the class, replacing the fold's exponent by $p = 1$, the class of the hazard models in use. To compare shapes rather than levels, I set the two prices equal when none of the budget has been spent, $\epsilon = 1$. Since the fold's price goes as $\epsilon^{-1/2}$ and the other stays constant, the fold's price is then larger by the factor $\epsilon^{-1/2}$, which grows as the budget is spent. Today this factor is largest at the lowest threshold of the range, where the least budget is left: it is 2.65 there. It reaches 6.1 at the dynamic scale ϵ_{dyn} , below which the budget drains faster than the circulation recovers and the power law stops.

Figure 4 plots the two factors against the proximity ϵ , the share of the warming budget still left before the threshold; spending the budget means moving from right to left. The flat line is the disagreement over location, the rising line the disagreement over class, and the hatched band shows where the circulation stands today across the published thresholds. The two lines cross at

$$\epsilon^* = \frac{\Delta T_{\text{low}}^* - \Delta T_0}{\Delta T_{\text{high}}^* - \Delta T_0} = 0.029, \quad (15)$$

just above $\epsilon_{\text{dyn}} = 0.027$. Over most of the window the location of the threshold, debated across

several degrees, moves the price more. The class, which the literature chooses implicitly, catches up near the end of the window, with about 2.9% of the budget left.

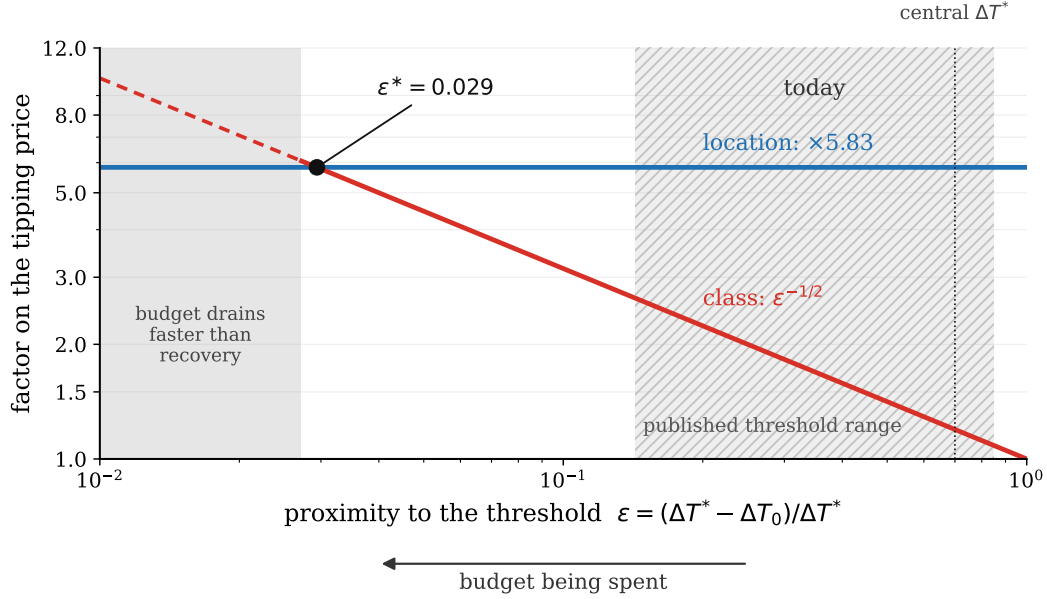


Figure 4: Which disagreement moves the tipping price more, for the overturning circulation. Horizontal axis: share of the warming budget left. Vertical axis: factor by which each disagreement multiplies the price. Shaded: below ϵ_{dyn} , where the power law stops. Log scales.

4.2 From the price to a deadline

Near a fold the price is steep and its level cannot be identified in time; policy can rely on a date instead: the date after which stopping emissions can no longer avoid the threshold. This section derives that date and the condition under which stopping by then is optimal; the next section computes it for the elements.

For a fold, (H1) gives $p = \frac{1}{2}$ and $c = 1$, and the noise layer is $\ell = \sigma^{2/3}$. Two changes adapt the price to a climate element. The budget is the carbon budget to the threshold, M^* , so the proximity is $\epsilon = \mu/M^*$. And the loss is not incurred at once: an element realizes its damage over a horizon H after crossing, decades for the AMOC and millennia for an ice sheet, which multiplies the loss by the average discount factor $\Omega(\rho H) = (1 - e^{-\rho H})/(\rho H)$. The price (11) then becomes

$$\text{SCC}_{\text{threshold}}(\epsilon) = \frac{A}{\sqrt{\epsilon}}, \quad (16)$$

where, by (11) with $p = \frac{1}{2}$ and $c = 1$, the level collects the loss, its realization factor, the steepness

and the budget,

$$A = \frac{\mathcal{L} \Omega(\rho H)}{2\sqrt{M^*}} \frac{\Phi(\tilde{\rho})}{\ell(\sigma)}. \quad (17)$$

The factor $\epsilon^{-1/2}$ is the shape, the same for every fold element; how the damage unfolds after crossing enters the level only. Anything that is a ratio in ϵ therefore comes from the shape alone and is available; a level in dollars per tonne needs A , which, as I show below, is not available.

The decision is when to stop drawing down the budget. Let the planner choose an abatement rate $z(t) \in [0, 1]$ at cost $\gamma(z)$ per unit of gross emissions E_0 , with $\gamma(z) = \Gamma(z)K/E_0$ the cost of Section 2.1 expressed per tonne, so that $\dot{M} = E_0(1 - z)$. Optimality equates the marginal abatement cost to the price, $\gamma'(z^*) = \text{SCC}_{\text{threshold}}(\epsilon)$; the Ramsey component adds to the right-hand side and brings the switch forward, so every date below is an upper bound on that account as well. Suppose a technology abates at a constant unit cost $\bar{\gamma}$. The marginal cost is then flat, and the planner goes all in or does nothing: abatement is full once the price exceeds $\bar{\gamma}$ and zero before. With (16) the switch is at

$$\frac{A}{\sqrt{\epsilon_\infty}} = \bar{\gamma} \implies \epsilon_\infty = \left(\frac{A}{\bar{\gamma}}\right)^2. \quad (18)$$

The switch exists only if the price reaches $\bar{\gamma}$ while the power law still holds, that is if $\epsilon_\infty \geq \epsilon_{\text{dyn}}$. If the damage is too small for that, the planner does not stop before the threshold, and the bound below is the date at which the threshold is reached rather than a date the planner meets. Before the switch the proximity falls at the rate E_0/M^* ; after it, the planner no longer emits and the proximity stops falling. Setting $\epsilon(t_c) = \epsilon_\infty$ gives the finite date $t_c = (M^*/E_0)(\epsilon_{\text{now}} - \epsilon_\infty)$. Along the optimal path the price never exceeds $\bar{\gamma}$: it tracks $A/\sqrt{\epsilon(t)}$ until the switch.

Two of the three quantities in t_c are observed: the proximity ϵ_{now} , from the threshold literature, and the ratio M^*/E_0 , from the budget and current emissions. The floor ϵ_∞ is not. It depends on the level A , which cannot be computed: the steepness Φ/ℓ needs the noise in the units of the model's state, and converting the noise observed in degrees or in a vegetation index into those units needs the distance between the two states in physical units, a quantity no dataset reports. And (18) squares A , so the floor responds to that missing distance more sharply than the price does. The floor is non-negative and enters with a minus sign, so dropping it turns the date into an upper bound,

$$t_c \leq \frac{M^*}{E_0} \epsilon_{\text{now}} = \frac{\mu_{\text{now}}}{E_0} = \frac{(\Delta T^* - \Delta T_0) \times 1,000}{\text{TCRE} \times E_0}, \quad (19)$$

where the third expression writes the budget as $M^* = \Delta T^* \times 1,000/\text{TCRE}$ in GtCO₂, with the TCRE in degrees per 1,000 GtCO₂. The bound is the date after which stopping emissions can no longer avoid the threshold. It needs neither the damage, nor the noise, nor the exponent, nor the class of the element, and since it drops the only quantity the price law reaches, it holds whether or not that law describes the element today. Before that date, stopping keeps the stable state in

existence: tipping remains possible, through the noise, but the earlier emissions stop the rarer and later it is. After that date, tipping is certain. A hazard model has no such date: no stock makes crossing certain, so stopping always lowers the risk and never ends it, and a bounded price can guide abatement on its own. Table 2 sets the bound against the three disagreements of the field: two of them, over the damage and over the class, leave the deadline where it is, and only the location of the threshold moves it, in proportion to the warming left before it.

Table 2: What each disagreement moves.

Disagreement over	Tipping price	Deadline bound (19)
Damage at tipping	its level, in proportion	nothing
Location of the threshold	$\times 5.83$ across the published range	in proportion to the warming left
Class (motion of the boundary)	$\times \epsilon^{-1/2}$, up to 6.1 for the AMOC	nothing

Why the instrument should be a quantity follows from the exponent. In the terms of [Weitzman \(1974\)](#), when the cost of abating is uncertain, a price is the better instrument if the marginal cost of abating rises faster in the quantity than the marginal benefit of restraint falls, and a quantity is the better instrument otherwise. For climate change as a whole the comparison has favoured prices, because the marginal damage of the stock is nearly flat over the quantities a policy moves ([Hoel and Karp, 2001](#); [Newell and Pizer, 2003](#)). Near a fold it is not. The marginal benefit of restraint is the price (16), and its slope in the stock is

$$\frac{d\text{SCC}_{\text{threshold}}}{dM} = \frac{A}{2M^*} \epsilon^{-3/2}, \quad (20)$$

which grows without bound as the budget runs out, while the slope of the marginal cost of abating, in the tonnes abated, does not depend on the proximity. Below the proximity at which the two slopes are equal, a quantity is the better instrument, for any finite slope of the marginal cost. Under the hazards in use, $p = 1$, the price tends to a constant near the threshold and its slope stays bounded, so they give no such presumption. Near a pitchfork the slope is unbounded too, but of the opposite sign: the price falls as the stock rises, so the marginal benefit of restraint rises with the tonnes abated, and the comparison favours a price. Where the proximity lies depends on the level A and is not identified, like the floor ϵ_∞ ; that it exists does not depend on A . The date of (19) is that quantity: a cap on cumulative emissions, stated as a time along a given emission path.

4.3 Which element comes first

I compute the bound (19) for the Amazon and the WAIS. It needs four inputs: the threshold, from [Armstrong McKay et al. \(2022\)](#); current warming, from [Forster et al. \(2024\)](#); the TCRE, from

IPCC (2021); and the path of emissions. The noise, the recovery rate and the damage enter only the price. The realization times used below are also those of Armstrong McKay et al. (2022), and Table 4 gives every input and its source. The threshold and the TCRE are uncertain, so I draw them from distributions fitted to their published ranges and report the median and the tenth percentile of the bound. I report it on three paths: emissions held at today's rate, $E_0 = 40 \text{ GtCO}_2$ per year (Friedlingstein et al., 2024); the middle-of-the-road path SSP2-4.5; and the high-emissions path SSP3-7.0 (Meinshausen et al., 2019). On a path whose emissions change over time, the bound is the year at which emissions summed from today reach the remaining budget μ_{now} .

The AMOC gets no date in the text. The geometry fixes the exponent and the damage fixes the level and the sign, and for the AMOC the sign is contested. Anthoff et al. (2016) find a small welfare gain from its slowdown, which Keen et al. (2022) dispute; a gain would reverse the price of restraint and leave the deadline without object. The losses of the Amazon and the WAIS carry no such ambiguity: their contributions to the social cost of carbon are positive in Dietz et al. (2021). Figure 5 shows the bounds for Amazon and WAIS. At today's emissions the WAIS comes first. Its median bound is 27 years, 25 on the middle-of-the-road path and 20 on the high-emissions path, and for a quarter of the draws the threshold already lies below current warming, so the bound is zero on every path⁸. The Amazon follows at 131 years at the median and under 54 in one draw in ten at today's emissions; on the middle-of-the-road path, whose emissions fall to zero by 2250 in its published extension, the threshold is not reached in 71.7% of draws: that path itself stops in time. On the high-emissions path its median bound is 78 years. The ice sheet's date barely depends on the path, from 20 to 27 years at the median; the Amazon's depends on it entirely.

Three conclusions follow. First, a ranking by price would reverse the order of action. The ice sheet, closest to its threshold, realizes its damage over millennia, so its loss is discounted most: the realization factor $\Omega(\rho H)$ is 0.017 for the ice sheet against 0.32 for the Amazon. What decides the order of action is the deadline, not the price.

Second, the cost of the technology that ends emissions changes when the planner stops, never the date. Cheap abatement makes the planner stop early, with part of the budget unspent; expensive abatement makes it stop closer to the threshold. However expensive abating is, a planner who stops at all does so before the date, since after it the element tips. Uncertainty about that cost is therefore not a reason to wait beyond it.

Third, the date says until when the threshold can be avoided, not whether avoiding it is worth its cost. That turns on how high the price can rise before the budget drains faster than the element recovers. Under the leading law, and with the state already inside the noise layer, that highest

⁸Armstrong McKay et al. (2022) report 1.5°C with a range of $[1.0, 3.0]$; current warming is 1.2°C (Forster et al., 2024). The share, 25.2% of the draws, is the centile of current warming in the distribution fitted to that published range, not a probability of collapse. Figures in the text are conditional on $t_c > 0$.

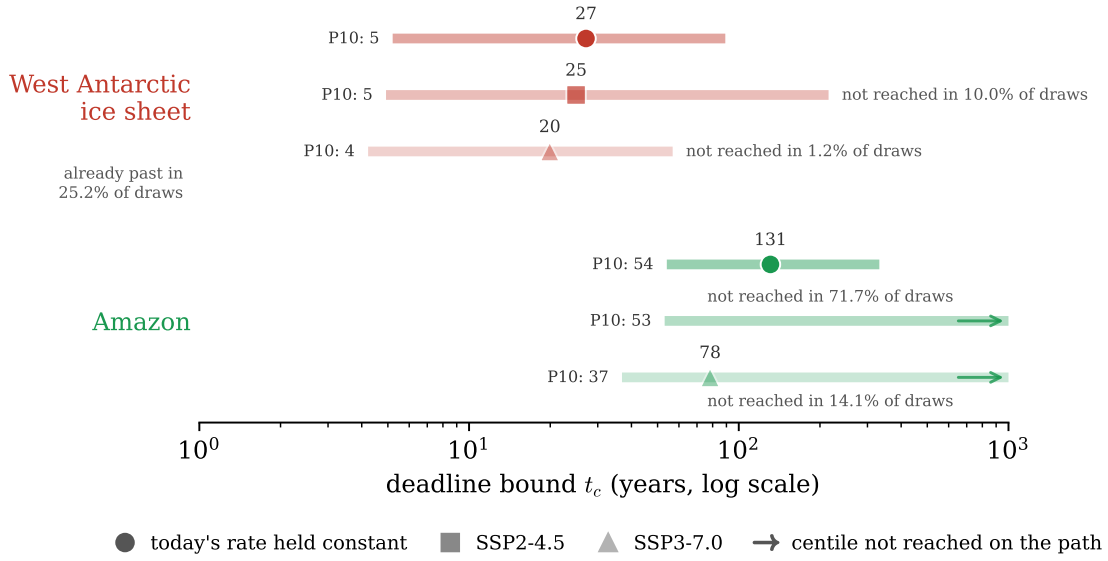


Figure 5: Deadline bound (19) for the Amazon and the WAIS on three emission paths. Markers: medians; bars: P10–P90 over the inputs of Table 4. Log scale. SSP2-4.5 brings emissions to zero by 2250, so the threshold may not be reached. The bound does not depend on the cost of abating.

price is $F = (\epsilon_{\text{now}}/\epsilon_{\text{dyn}})^{1/2}$ times today's, and stopping in time is optimal only if it exceeds the cost of abating. For the Amazon, F is about 10.8 if the forest recovers at the rate Boulton et al. (2022) measure for its vegetation index, and close to one if it recovers over a century. For the West Antarctic ice sheet, whose flow relaxes over 10 to 20 years (Hill et al., 2023), with slower modes acting over centuries or longer (Robel et al., 2018), it is at most 1.2.

5 Conclusion

When a controllable stock drifts toward a point of no return, what is a unit of restraint worth? Economics answers through the probability of crossing. This paper shows that the probability is not enough: the price of restraint depends on how fast the boundary moves toward the state as the budget is spent, a rate that the hazard leaves free and that the record cannot reveal before most of the budget is gone. The dynamics of the system supply it. At a fold, the form the physical literature reports for the overturning circulation, the West Antarctic ice sheet and the Amazon, the boundary moves as the square root of the budget left, and the price rises as its inverse square root until the budget drains faster than the system recovers. But the price grows large only once crossing has become likely, so a price signal arrives late.

Two questions surround a tipping threshold: whether avoiding it is worth its cost, and until

when it can still be avoided. The first needs the damage, and the damage is hard to value: much of it lies outside markets, in the rainfall that a weaker overturning circulation would shift or the species that would go with the Amazon forest, and the estimates in use are disputed (Keen et al., 2022; Dietz et al., 2022). This paper states the condition on which that question turns but leaves it open. The second needs neither the damage nor the price: the budget to the threshold, divided by the rate of emissions, gives the date after which stopping emissions can no longer avoid the threshold. On the high-emissions path SSP3-7.0, that date is 78 years away for the Amazon at the median, and 37 years or less in one draw in ten; the threshold of the West Antarctic ice sheet already lies below current warming in 25.2% of draws. These are latest dates. Counting greenhouse gases other than CO₂ would, in most scenarios, bring them forward, and so would interactions between elements: a reinforcing coupling lowers a threshold, and a neighbour that tips first cuts the budget left at once. Whatever the answer to the first question, it has to be given before that date. A tipping threshold then calls for a constraint on cumulative emissions, as remaining carbon budgets already are for temperature targets, as a complement to a premium on the price of carbon. The probability of crossing, taken as a primitive, leaves the price near a point of no return open; the budget left to the threshold, divided by the pace at which it is spent, gives the date by which to act without it.

The argument rests on a condition rather than on a domain: a controllable stock drifts toward a boundary, and the system's own dynamics fix how the budget moves that boundary. Aggregate leverage drifting toward default meets it, with a better record, since the distance to default is measured and one could ask whether a fire-sale feedback makes the default boundary move toward the firm (Brunnermeier and Sannikov, 2014); so does a harvested population drawn toward the stock below which it does not recover. A risk whose probability the decision-maker does not move lies outside it, and there the classical treatment is the right one.

Four limits bound the result. It is derived for a single slow variable; it survives for two weakly coupled folds, but larger networks, and spatially extended systems such as permafrost, may belong to a different problem. It holds while the budget drains slowly relative to the system's recovery. It does not treat tipping caused by the pace of the forcing rather than its level, which Ashwin et al. (2012) call rate-induced. And it assumes that policy moves the drift of the state, not its volatility.

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A Derivations

A.1 The amplitude (H3)

Write $\varphi = G u$ in (2). The terms in u' cancel when $\sigma^2 G' + fG = 0$, that is $G = e^{-B/\sigma^2}$ with $B' = f$, and dividing through leaves

$$u'' = \frac{2}{\sigma^2} Q u, \quad Q = \rho + \frac{f'}{2} + \frac{f^2}{2\sigma^2}, \quad (21)$$

in which the discount rate enters Q as an additive constant. At the merged point the drift is $-x^k$, taking $c_k = 1$, which a rescaling of the state allows, and with $x = \ell\eta$ as in (10) the equation becomes

$$u''(\eta) = 2\hat{Q}(\eta) u(\eta), \quad 2\hat{Q} = 2\tilde{\rho} - k\eta^{k-1} + \eta^{2k}. \quad (22)$$

Its decaying solution is unique up to a factor, and the amplitude is the rate at which it falls, per layer width, at the merged point,

$$\Phi(\tilde{\rho}) \equiv -\frac{u'(0; \tilde{\rho})}{u(0; \tilde{\rho})}. \quad (23)$$

At $\tilde{\rho} = 0$ the decaying solution is $u_0 = e^{-\eta^{k+1}/(k+1)}$, whose slope vanishes at the origin, so $\Phi(0) = 0$: restraint, which buys only delay, is worth nothing to a planner who does not discount. Let $\dot{u}$ be the derivative of u in $\tilde{\rho}$. Since $\tilde{\rho}$ enters $\hat{Q}$ additively, $\dot{u}'' = 2\hat{Q} \dot{u} + 2u$. Multiplying (22) by $\dot{u}$, this equation by u , and subtracting eliminates $\hat{Q}$: $(u' \dot{u} - \dot{u}' u)' = -2u^2$. The left-hand expression vanishes at infinity, because u decays faster than any power, so integrating outward from the origin gives

$$\Phi'(\tilde{\rho}) = \frac{2}{u(0; \tilde{\rho})^2} \int_0^\infty u(\eta; \tilde{\rho})^2 d\eta > 0. \quad (24)$$

With $\Phi(0) = 0$, the amplitude is positive and strictly increasing for every $\tilde{\rho} > 0$ and every order k , which is (H3).

A.2 Numerical methods

I solve the boundary-value problem (2) with a fourth-order collocation scheme on an adaptive mesh, on a domain that extends far enough beyond the stable state that moving its edge does not change the solution. The solver reproduces the scaling of (H2): at fixed $\tilde{\rho}$, the scaled shortfall varies across the grid of (σ, ρ) by at most 0.0038%, and it matches the amplitude of Appendix A.1 to within 0.0064%. The price is the slope $-\partial\varphi/\partial\epsilon$ at a fixed state $x^*(\epsilon)$, computed by centred finite differences on the numerical solution, and its local slope in $\log \epsilon$ is read on the same grid. The error of the leading law falls faster than the square root of the budget: a regression of its logarithm on

$\log \epsilon$, for $\epsilon < 0.01$, gives a slope of 0.62, with a 95% interval from 0.61 to 0.63.

A.3 The price while the budget is spent

The paper computes the price with the budget frozen while the state moves; here the budget drains at constant emissions. The draining crossing value solves, for the fold,

$$E_0 \frac{\partial \varphi}{\partial \mu} = \frac{\sigma^2}{2} \varphi_{xx} + f \varphi_x - \rho \varphi, \quad \varphi(x_-(\mu), \mu) = 1, \quad \varphi(x, 0) = 1,$$

the last condition saying that crossing is immediate once the budget is exhausted. The equation is parabolic forward in μ , so it is solved by an implicit scheme, advancing in μ from zero, in the distance to the boundary, which holds the boundary in place. The draining price is the derivative of this solution at a fixed state, and the frozen price the same derivative of the problem without the left-hand side, on the same grid. Every number is computed twice, the second time with both grids doubled; the computation stops if the doubling moves the price beyond its tolerance, and the numbers reported come from the doubled grids. For the overturning circulation, the draining price begins to depart from the frozen one between 0.2 and 1.7 times ϵ_{dyn} , depending on where the state sits; below ϵ_{dyn} it first rises faster, by a factor of up to 1.8, then flattens. For the West Antarctic ice sheet, ϵ_{dyn} lies close to today's proximity, so the paper uses only its date.

B What the record can establish

The calculations of Section 2.4 are made on the record as it is observed, 146 annual means of a continuous process, simulated with the recovery rate θ set to reproduce the lag-one autocorrelation of the detrended annual series. Under the alternative, the recovery rate falls as a power of the time left before the threshold at constant emissions, $\theta \propto t_r^a$, where t_r falls by one year every year, with $a = \frac{1}{2}$ for a fold and $a = 1$ for a transcritical.⁹ Under the null, the recovery rate is constant and the variance of the noise drifts along a one-parameter family of paths. For each alternative I find the least favourable member of the family and compute the power of the most powerful test against it, which, by the Neyman–Pearson lemma, no test, whatever its statistic, exceeds. Today's distance, $t_r = 156$ years, is the deadline bound (19) at central inputs. The least favourable member is located on a grid of the family's parameter, and the minimum of power estimates taken over a grid inherits their Monte Carlo spread: on the three grids computed, the bound for a fold today runs from 0.11 to 0.13, and the text reports the finest. The calculations of Ditlevsen and Ditlevsen (2023) use

⁹In general $a = p_d(k - 1)$, with p_d the exponent at which the margin closes, $d \propto \mu^{p_d}$, and k the order of the drift at the merged point.

monthly anomalies; everything here is on annual means.

C Calibration

C.1 Where the elements sit

The dichotomy (11) holds on the fixed-budget regime, bounded below where the budget no longer stands still while the state crosses and above where the stable state leaves the noise layer. None of the three elements is inside it today. The upper edge is not computed, but an argument places the overturning circulation and the Amazon above it: an element close to its edge, measured against its own fluctuations, is carried across within a few recovery times, and both have held their state far longer.

The lower edge of the fixed-budget regime is the proximity ϵ_{dyn} at which the budget, in relative terms, drains as fast as the element recovers. Near a fold the recovery rate falls as the square root of the remaining budget, $\theta(\epsilon) = \theta (\epsilon/\epsilon_{\text{now}})^{1/2}$, while the budget drains at the relative rate $E_0/(M^*\epsilon)$; equating the two gives

$$\epsilon_{\text{dyn}} = \left(\frac{E_0 \epsilon_{\text{now}}^{1/2}}{M^* \theta} \right)^{2/3}.$$

Below ϵ_{dyn} the boundary moves while the state approaches, and the price leaves the power law (Section 3.3.3, Appendix A.3). Between today and ϵ_{dyn} the leading law therefore lifts the price of restraint by $F = (\epsilon_{\text{now}}/\epsilon_{\text{dyn}})^{1/2}$. F measures the width of the window.

It is the lift of the full price only if the state is already inside the noise layer today. The full solution of (2) shows how much this matters for the overturning circulation at its measured recovery rate. If the state sits deep in the layer today, at $\eta^* = 0.1$, the lift is 11% smaller than F . If it sits at the edge of the layer, at $\eta^* = 1$, the lift is 87% smaller. There the crossing value falls fastest as the margin opens, so the price is already close to its level at ϵ_{dyn} . Further out the crossing is rare, today's price is exponentially small, and the lift exceeds F many times over. Today's η^* is not identified, because it requires the physical margin. Whether the fixed-budget regime is reached before ϵ_{dyn} depends on it: η^* falls as the square root of the proximity, so at ϵ_{dyn} it is η_{now}^*/F , and the law describes the price there only if that value is small (Section 3.3.1). I therefore report F as the width of the window, not as a forecast of the rise.

Table 3 reports ϵ_{dyn} and F . Only the overturning circulation has a measured recovery rate, 0.79 per year from its record, which gives $F = 5.1$ at today's emissions. For the Amazon, Boulton et al. (2022) measure a recovery rate of 9.68 per year for the vegetation index; if the forest recovered at that rate, F would be 10.8, and if it recovered over a century, 1.1. For the West Antarctic ice sheet, whose flow relaxes over 10 to 20 years (Hill et al., 2023), with slower modes acting over centuries

or longer (Robel et al., 2018), F is at most 1.2, and the window is empty for the slower modes, so the conclusion holds across both sources: the ice sheet is already close to its threshold and the budget outruns it. Proximity falls as the budget is spent, so an element outside the fixed-budget regime can enter it before its deadline; what the paper prices is the regime these elements can occupy. The deadline bound of Section 4.2 and the ranking of Section 4.3 do not pass through the fixed-budget regime at all.

Table 3: Dynamic scale ϵ_{dyn} and the factor F by which the leading law lifts the price of restraint between today’s proximity and ϵ_{dyn} .

Element	Recovery rate θ (per year)	ϵ_{now}	Today’s rate		SSP2-4.5		SSP3-7.0	
			ϵ_{dyn}	F	ϵ_{dyn}	F	ϵ_{dyn}	F
AMOC	0.79, record	0.70	0.027	5.1	0.029	4.9	0.044	4.0
Amazon	9.68, vegetation index	0.66	0.0056	10.8	0.0060	10.5	0.0091	8.5
Amazon	one per century, assumed	0.66	0.55	1.1	0.59	1.1	0.89	—
West Antarctic ice sheet	1/10, flow	0.20	0.13	1.2	0.14	1.2	0.17	1.1
West Antarctic ice sheet	1/20, flow	0.20	0.20	—	0.22	—	0.27	—
West Antarctic ice sheet	one per century, slower modes	0.20	0.60	—	0.64	—	0.78	—

$F = (\epsilon_{\text{now}}/\epsilon_{\text{dyn}})^{1/2}$; — where $\epsilon_{\text{dyn}} \geq \epsilon_{\text{now}}$, so that the window is empty. M^* is the Monte Carlo median of the budget, 9433 GtCO₂ for the AMOC, and ϵ_{now} the proximity at central inputs. On SSP2-4.5 and SSP3-7.0, E_0 is replaced by the highest annual emissions met before the median date of the bound on that path, or over the whole path when the median is not reached. Recovery rates: the overturning circulation from its record (Appendix C.2); the Amazon from the vegetation index of Boulton et al. (2022), or one per century as a sensitivity; the ice sheet from the relaxation of its flow (Hill et al., 2023), or one per century for its slower modes (Robel et al., 2018).

C.2 The inputs

The recovery rate of the AMOC is read from the lag-one autocorrelation of the detrended annual HadISST series of Caesar et al. (2018) (1871–2016), as a property of the continuous process whose annual means the series records; read as a point sample, it would be understated by a factor of 1.27.

The thresholds ΔT^* and the realization times H are the central estimates and ranges reported by Armstrong McKay et al. (2022), and current warming, the observed mean over 2014–2023 (Forster et al., 2024), is $\Delta T_0 = 1.2^\circ\text{C}$. Warming is converted into emissions through the IPCC AR6 climate response to cumulative emissions, drawn in degrees per 1,000 GtCO₂ from a normal distribution with mean 0.45 and standard deviation 0.19, so that its 17th and 83rd percentiles reproduce the likely range of 0.27 to 0.63 (IPCC, 2021, SPM D.1.1); draws below zero, 0.9% of the total, are discarded. The budgets count CO₂ only. In most scenarios, warming from other forcings adds to the warming from CO₂ (IPCC, 2021, SPM D.1.7 and Section 5.5.2.2.3); counting it would bring each

date forward, so the dates remain latest dates. The budget to the threshold is

$$M^* = \frac{\Delta T^*}{\text{TCRE}} \times 1,000 \text{ GtCO}_2, \quad (25)$$

the part still unspent is $\mu_{\text{now}} = M^* \epsilon_{\text{now}}$, and the proximity reduces to a ratio of temperatures, $\epsilon_{\text{now}} = \max(0, (\Delta T^* - \Delta T_0)/\Delta T^*)$; for the AMOC, $(4 - 1.2)/4 = 0.70$. I set $\rho = 0.03$ and draw $N = 100,000$ Monte Carlo samples, reporting medians and 80% intervals.

The two emission paths of Section 4.2 are the global CO₂ emissions, fossil and land use, of SSP2-4.5 and SSP3-7.0 with their extensions beyond the end of the century (Meinshausen et al., 2019), as published by the RCMIP protocol (Nicholls et al., 2020), which reports them at intervals of up to a decade; I interpolate linearly between the years it reports. The draws of the threshold and the transient response are those of the constant path, so the three bounds differ only through the path along which the remaining budget is spent.

Table 4: Calibration inputs for the elements. Central value with [P10, P90]; M^* is the Monte Carlo median of the budget; ϵ_{now} is the current proximity at central inputs. The ranges are the published ranges of Armstrong McKay et al. (2022); the Monte Carlo fits a log-normal law to each, with the median at the central value and a spread set by the width of the published range.

Element	ΔT^* (°C)	M^* (GtCO ₂)	H (yr)	ϵ_{now}
AMOC	4 [‡] [1.4, 8.0]	9433	50 [15, 300]	0.70
Amazon	3.5 [2.0, 6.0]	7998	100 [50, 200]	0.66
WAIS	1.5 [1.0, 3.0]	3895	2000 [500, 13,000]	0.20

[‡] Across the published range of AMOC thresholds, today’s proximity runs from 0.143 to 0.850. I calibrate at the central threshold of Armstrong McKay et al. (2022).

D Two coupled elements

Tipping elements interact: for instance, a weaker overturning circulation changes rainfall over the Amazon. The literature prices the interaction in integrated models (Lemoine and Traeger, 2016). This appendix asks whether such an interaction changes the exponent p of Section 2.2. It does not, as long as the coupling is weak and the second element has not tipped.

Let X_1 and X_2 be the states of two elements. The drift of the first depends on the state of the second:

$$dX_1 = [f_1(X_1; M) + \beta \xi(X_1, X_2)] dt + \sigma_1 dW_1,$$

where f_1 is the drift of the first element on its own, ξ is a smooth function that describes how the second element acts on the first, the real number β measures the strength of the coupling, σ_1 is the

noise of the first element and W_1 is a standard Brownian motion. Suppose the second element stays near its stable state $\bar{X}_2(M)$, which moves smoothly with the stock M . Its fluctuations around that state enter the first equation as an additional, state-dependent noise; I neglect them, which leaves the geometry unchanged. The drift of the first element is then

$$\tilde{f}_1(X_1; M, \beta) = f_1(X_1; M) + \beta \xi(X_1, \bar{X}_2(M)).$$

When $\beta = 0$ the first element has a fold at the stock M_1^* and the state x_c : there $\tilde{f}_1 = 0$ and $\partial \tilde{f}_1 / \partial X_1 = 0$. The fold is non-degenerate and transversal: $\partial^2 \tilde{f}_1 / \partial X_1^2 \neq 0$ and $\partial \tilde{f}_1 / \partial M \neq 0$. The fold is a solution (X_1, M) of the two equations $\tilde{f}_1 = 0$ and $\partial \tilde{f}_1 / \partial X_1 = 0$, and the two conditions make the matrix of their derivatives in (X_1, M) invertible, with determinant $-(\partial \tilde{f}_1 / \partial M)(\partial^2 \tilde{f}_1 / \partial X_1^2) \neq 0$. By the implicit function theorem, the two equations still have a solution for β small enough, close to the uncoupled one and moving smoothly with β , at which the two conditions still hold. The coupled element therefore has a fold, at a shifted threshold $M_1^*(\beta)$. The boundary then moves as the square root of the budget left to the shifted threshold, $\mu_1 = M_1^*(\beta) - M$, and the exponent stays $p = \frac{1}{2}$. The coupling moves the threshold and changes the constant in front of the price. It does not change the class.

The direction of the shift follows from the same two equations. Along the fold $\partial \tilde{f}_1 / \partial X_1 = 0$, so differentiating $\tilde{f}_1 = 0$ with respect to β gives $(\partial \tilde{f}_1 / \partial M) dM_1^* / d\beta + \partial \tilde{f}_1 / \partial \beta = 0$. At $\beta = 0$ this is

$$\frac{dM_1^*}{d\beta} = - \frac{\xi(x_c, \bar{X}_2(M_1^*))}{\partial f_1 / \partial M}.$$

Call the coupling reinforcing when $\beta \xi(x_c, \bar{X}_2(M_1^*))$ has the sign of $\partial f_1 / \partial M$, that is, when the second element pushes the first in the same direction as more emissions. For β small enough, a reinforcing coupling lowers the threshold: $M_1^*(\beta) < M_1^*$.

Two consequences follow. First, a coupled element must be priced against its coupled threshold. Measured against the uncoupled threshold M_1^* , the budget is off by $M_1^*(\beta) - M_1^*$. Near the threshold this error is as large as the budget itself, and the price would not appear to follow the power $\frac{1}{2}$. Under a reinforcing coupling the uncoupled budget overstates the margin, so close to the coupled threshold the uncoupled price understates the price. Second, the argument requires the second element to stay near its stable state. If it tips first, $\bar{X}_2$ jumps, and the drift of the first element jumps with it. So does its threshold, and the budget left to it falls at once. This is a cascade. Its price adds a term for the jump to the fold term, and this paper does not analyse it.